

Notes: Huang & Kisgen (JFE, 2013)

Gender and Corporate Finance: Are Male Executives Overconfident Relative to Female Executives?

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1 Big picture

- **Question.** The paper asks whether executive gender is associated with different corporate investment and financing choices, and whether those differences are consistent with male executives being more overconfident than female executives.
- **Core hypothesis.** Prior psychology and finance evidence suggests men are relatively more overconfident. In a corporate finance setting, overconfident managers overestimate project quality or underestimate uncertainty. They therefore undertake more marginal acquisitions and financing decisions, including some negative-NPV actions.
- **Main setting.** The authors hand collect a sample of CEO and CFO transitions from 1994–2003, focusing on male-to-female transitions and comparing them with male-to-male transitions. The sample is constructed from ExecuComp plus manual SEC filing searches for firms with book assets above \$500 million.
- **Empirical design.** The main tests use a difference-in-differences framework around executive transitions. The treatment group is firms that switch from a male executive to a female executive. The control group is firms that switch from a male executive to another male executive.
- **Main decision result.** Firms with female executives grow more slowly, make fewer acquisitions, and issue debt less often. Equity issuance is not robustly different.
- **Main value result.** The market reacts more favorably to acquisitions and debt issues when the firm has a female executive. Acquisition announcement returns are roughly 2 percentage points higher, and debt-issue announcement returns are about 3.7–5.4 percentage points higher depending on the return measure.
- **Direct overconfidence evidence.** Male executives issue narrower earnings forecasts, are less likely to exercise deep-in-the-money options early, and are more likely to undertake acquisitions with negative announcement returns.
- **Risk-aversion alternative.** A risk-aversion explanation predicts women may avoid both bad and good projects. But it does not predict that male-led acquisitions are more likely to destroy value, nor that men have worse career outcomes. The paper uses these tests to argue for relative male overconfidence rather than female risk aversion alone.
- **Identification concern.** Female executives are not randomly assigned to firms. The paper addresses this using firm fixed effects around transitions, controls, propensity-score matching, and an IV based on state-level gender status equality.
- **Takeaway.** Managerial gender is associated with economically meaningful differences in corporate decisions. Models that explain acquisitions and capital structure only with firm characteristics miss an important managerial characteristic.

2 Data and measurement

2.1 Executive transition sample

- The sample period is 1993–2005 for executive identification, with transitions from 1994–2003.
- CEOs and CFOs are both included because prior research shows both positions matter for investment and financing choices.
- The initial executive list comes from ExecuComp. Since ExecuComp covers large firms, the authors supplement it with hand collection from SEC 10-K filings for all firms with book assets above \$500 million.
- Firm financial data come from Compustat. The firm must be listed on NYSE, Amex, or Nasdaq.
- The authors collect the executive’s name, gender, and rank. If a filing does not identify gender, they search Factiva, company websites, Forbes, ZoomInfo, and related sources.

- The final transition sample requires the new executive to appear as CEO or CFO for at least three consecutive years.
- The comparison is mainly male-to-female transitions versus male-to-male transitions. This makes the design a transition-based comparison rather than a simple cross-sectional comparison of female and male executives.

Interpretation: The sample is deliberately designed around executive turnover so that the authors can compare the same firm before and after a change in executive gender, while using male-to-male transitions as a control for normal turnover dynamics.

2.2 Why CEOs and CFOs are both included

- CEOs influence strategy, acquisitions, and broad investment policy.
- CFOs influence financing, capital structure, acquisition execution, and financial policy.
- The paper emphasizes that CFOs are especially important for capital structure and acquisition processes.
- In the sample, female executives are more often CFOs than CEOs because female CFOs were more common than female CEOs during the sample period.

Interpretation: Pooling CEOs and CFOs increases statistical power and focuses on the executives most closely involved in the acquisition and financing decisions studied in the paper.

2.3 Main decision variables

The paper studies several firm decisions:

- **Asset growth:** percentage change in total assets.
- **Acquisitions:** acquisitions divided by net property, plant, and equipment.
- **Book leverage:** debt divided by debt plus book equity and related balance-sheet claims.
- **Market leverage:** debt divided by debt plus market equity and related balance-sheet claims.
- **Debt issuance:** log of long-term debt issuance.
- **Equity issuance:** log of sales of common and preferred stock.

Interpretation: The acquisition and debt issuance variables are the closest direct tests of the overconfidence story. If men overestimate project value or their own precision, they should engage in more significant external actions.

2.4 Announcement returns

- The paper measures three-day cumulative abnormal returns around announcements of acquisitions, debt issuance, and equity issuance.
- It reports both raw CARs and market-adjusted CARs.
- For acquisitions, controls include deal features such as stock deal, public target, diversifying deal, hostile offer, toehold, contested bid, tender offer, and merger of equals.
- For security issuance, controls include firm characteristics and market microstructure variables such as return volatility, share turnover, and log proceeds.

Interpretation: Announcement returns reveal whether the market views female-led and male-led transactions differently. Under male overconfidence, male-led transactions should have lower CARs because some transactions are negative NPV.

2.5 Additional overconfidence measures

The paper uses four additional tests to distinguish male overconfidence from female risk aversion:

- **Earnings forecast spread.** Overconfident executives should issue narrower forecast ranges.
- **Career outcomes.** If men make value-destroying decisions, male executives should be more likely to leave or be replaced.
- **Option exercise and stock purchase behavior.** Following Malmendier and Tate, overconfident executives delay option exercise and increase exposure to firm-specific risk.
- **Value-destroying acquisitions.** Male-led firms should be more likely to announce acquisitions with negative abnormal returns.

Interpretation: These tests are important because the main action-frequency results alone could also be explained by female risk aversion. The direct overconfidence tests help pin down the behavioral interpretation.

3 Theory, hypotheses, and empirical specifications

3.1 Behavioral model of overconfidence

Consider a project that costs \$1 today and pays expected value a in one year, with discount rate s . A rational executive accepts if the discounted payoff exceeds cost. An overconfident executive may:

- overestimate a , the expected payoff;
- underestimate uncertainty or discount risk too lightly;
- believe his own forecast is more precise than it is.

Interpretation: If male executives are relatively overconfident, they expand the set of projects they perceive as positive NPV. They will therefore undertake more acquisitions and debt-financed actions, but the marginal actions are more likely to disappoint shareholders.

3.2 Prediction 1: frequency of corporate actions

$$\Pr(\text{Acquisition or debt issue} \mid \text{male executive}) > \Pr(\text{Acquisition or debt issue} \mid \text{female executive}). \quad (1)$$

Interpretation: Male overconfidence predicts more action, especially actions with discretion and large managerial judgment. The paper therefore focuses on acquisitions and debt issuance.

3.3 Prediction 2: announcement returns

$$E[CAR \mid \text{female executive transaction}] > E[CAR \mid \text{male executive transaction}]. \quad (2)$$

Interpretation: If male executives take more marginal negative-NPV projects, market reactions to male-led transactions should be lower. Female-led transactions should be selected from a stricter acceptance threshold.

3.4 Prediction 3: earnings forecast ranges

The paper defines forecast spread as

$$ForecastSpread = \frac{UpperBound - LowerBound}{Midpoint}. \quad (3)$$

For point forecasts, the spread is zero.

Interpretation: Overconfidence implies overprecision. If men are more overconfident, male-led firms should issue narrower earnings forecast ranges than female-led firms.

3.5 Main difference-in-differences specification

For decision outcomes, the main regression is

$$Y_{i,t+1} = \mu + \nu_i + \tau_t + \beta_1 Post_{i,t+1} + \beta_2 (Female_i \times Post_{i,t+1}) + \theta X_{i,t} + \varepsilon_{i,t}. \quad (4)$$

- $Y_{i,t+1}$ is the decision variable: asset growth, acquisitions, leverage, debt issuance, or equity issuance.
- ν_i are firm fixed effects.
- τ_t are year fixed effects.
- $Female_i$ equals one for male-to-female transition firms.
- $Post_{i,t+1}$ equals one after the executive transition.
- β_2 is the difference-in-differences estimate.

Interpretation: The coefficient β_2 asks whether a firm behaves differently after switching to a female executive, relative to the change observed around male-to-male executive transitions.

3.6 Announcement-return specification

For transaction returns, the paper estimates

$$CAR_{i,t} = \mu + \tau_t + \beta_0 Female_i + \beta_1 Post_{i,t} + \beta_2 (Female_i \times Post_{i,t}) + \theta X_{i,t} + \varepsilon_{i,t}. \quad (5)$$

Interpretation: The outcome is the announcement CAR for an acquisition or security issuance. The coefficient on $Female \times Post$ measures whether market reactions improve after a male-to-female transition relative to male-to-male transitions.

3.7 Instrumental-variable specification

The instrument is the state-level gender status equality index of Sugarman and Straus, assigned using the firm's headquarters state.

The first stage is

$$Female_i = \phi + \tau_t + \gamma GenderEquality_i + \theta X_{i,t} + \eta_{i,t}, \quad (6)$$

and the second stage is

$$Y_{i,t} = \alpha + \tau_t + \beta \widehat{Female_i} + \theta X_{i,t} + \varepsilon_{i,t}. \quad (7)$$

Interpretation: The exclusion restriction is that state gender equality affects acquisition behavior only through the likelihood that a firm has a female executive. The paper reports first-stage significance, but also notes that the F-statistics are below the usual rule-of-thumb threshold of 10, so weak-instrument concerns cannot be fully ruled out.

4 Identification and empirical design

4.1 Why endogeneity is a concern

- Female executives are not randomly assigned.
- Boards may discriminate, which could create selection into particular firms.
- Women may self-select into certain industries, firms, or cultures.
- Firms anticipating lower acquisition activity may deliberately hire a female executive.
- Female executives are more common in consumer-oriented industries and in later years of the sample.

Interpretation: A simple cross-sectional comparison would not be credible. The transition design is meant to separate executive gender effects from fixed firm traits.

4.2 What the transition design controls for

- Firm fixed effects absorb time-invariant firm culture, industry, location, and historical acquisition policy.
- Year fixed effects absorb aggregate merger waves, credit conditions, and economy-wide changes in female executive representation.
- Male-to-male transitions provide a turnover control group, separating gender changes from generic management turnover effects.
- Excluding the transition year reduces ambiguity about which executive made the decision.

Interpretation: The identifying assumption is that, absent the female transition, male-to-female firms would have changed similarly to male-to-male transition firms.

4.3 Robustness designs

The paper uses three complementary designs:

1. **Difference-in-differences around transitions.** This is the preferred specification.
2. **Panel regressions with firm fixed effects.** These include more firms, including female-to-female or male-to-female-to-male transitions.
3. **Propensity score matching.** Treated observations are matched to comparable firms based on profitability, size, market-to-book, PPE, year, industry, and CEO indicator.

Interpretation: The main acquisition and debt issuance results persist across these designs, which strengthens the case that they are not only artifacts of one specification.

4.4 Interpretive limits

- The paper cannot prove that gender biologically causes the behavior; gender may proxy for selection, training, socialization, or labor-market discrimination.
- Most female executives in the sample are CFOs, so results are not purely about CEOs.
- The IV strategy is useful but not perfect because first-stage F-statistics are below 10.
- Female executive representation is small, limiting power in some subsample tests.
- The paper measures important corporate actions, but not all dimensions of executive performance.

Interpretation: The strongest conclusion is that major corporate decisions differ significantly after male-to-female executive transitions and that the pattern is most consistent with relative male overconfidence.

5 Tables and figures

5.1 Tables 1–2: sample structure and executive/firm characteristics

Read:

- Table 1 shows 1,750 male executives and 116 female executives in the transition sample.
- Female representation rises over time: 43.1% of female executives in the sample were hired from 2001–2003, compared with 37.6% of male executives.
- Female executives are more represented in consumer industries: 32.8% of female executives versus 25.5% of male executives.
- Table 2 shows female-transition firms are larger: at $t - 1$, average size is 9,177 for male-to-female firms versus 6,229 for male-to-male firms.
- Female executives are younger on average: 44.12 years versus 47.21 years.
- Female executives are more often hired internally: 64.29% versus 43.19%.
- Female representation on the board is higher in female-executive firms: 14.08% versus 10.22%.

Economic message: These tables show why the paper cannot rely on a simple female-versus-male comparison. Female executives appear in different years, industries, firm sizes, and governance environments, motivating the transition-based identification design.

Table 1

Summary statistics.

This table presents the distribution of our sample executives by gender and other characteristics, including tenure, transition year, and industry affiliation. Tenure is the number of years that the executive shows up on a firm's 10K reports as a Chief Executive Officer or Chief Financial Officer. We require that the executive has to be in office consecutively for at least three years. The year of transition is the first year that the executive shows up on the annual report. The industry definition follows Fama and French classification, which is available on Kenneth French's website (http://mba.tuck.dartmouth.edu/pages/faculty/ken_french/data_library.html).

Panel A: Distribution of executives by gender and tenure

Gender	Tenure (years)												Total
	3	4	5	6	7	8	9	10	11	12	10	12	
Male	401	347	321	246	158	124	62	49	32	10	1,750		
	22.9%	19.8%	18.3%	14.1%	9.0%	7.1%	3.5%	2.8%	1.8%	0.6%			
Female	21	36	21	14	12	6	3	1	2	0	116		
	18.1%	31.0%	18.1%	12.1%	10.3%	5.2%	2.6%	0.9%	1.7%	0.0%			

Panel B: Distribution of executives by gender and transition year

Gender	Transition Year											Total
	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2003	
Male	41	120	135	138	183	201	274	241	217	200	1,750	
	2.3%	6.9%	7.7%	7.9%	10.5%	11.5%	15.7%	13.8%	12.4%	11.4%		
Female	4	9	3	8	10	17	15	20	21	9	116	
	3.5%	7.8%	2.6%	6.9%	8.6%	14.7%	12.9%	17.2%	18.1%	7.8%		

Panel C: Distribution of executives by gender and industry affiliation

Gender	Industry					Total
	Consumer	Manufacture	Health	High-Tech	Other	
Male	1,150	450	100	100	100	1,750
Female	100	100	100	100	100	116

(a) Table 1: sample executives by gender, tenure, year, and industry.

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Table 2

Firm leverage, profitability, and size around executive transitions.

Panel A presents the mean statistics of leverage, profitability, and size of the sample firms around the year of transition (Year t). See Appendix B for the definition of the variables. Panel B reports board characteristics of male and female executive firms and executive characteristics. Board Size is the number of directors on the board. Board Independence is the fraction of independent outside directors on the board. CEO/Chairman Duality is an indicator variable that equals one if the titles of Chief Executive Officer (CEO) and chairman are vested in the same individual and zero otherwise. Female Representation on Board is the fraction of female directors on the board. MBA is an indicator variable that equals one if the executive has an MBA degree and zero otherwise. Executive Age is the age of the executive when hired as a CEO or Chief Financial Officer (CFO). Hired from Inside is an indicator variable that equals one if the executive is hired from inside the firm and zero otherwise. Total Compensation is the total amount of annual compensation the executive receives. Cash Compensation and Stock and Option Compensation are cash compensation (including salary and bonus) and stock and option compensation as a fraction of total compensation, respectively. The last two columns in Panel B report t -statistics and z -statistics for differences in means and medians, respectively, of each variable across male and female executive firms. Significance on a 10% (*), 5% (**), or 1% level (***) is indicated.

Panel A: Firm leverage, profitability, and size around executive transitions.

Year	Book Leverage		Market Leverage		Return on Assets		Operating ROA		Size	
	Male-to-male	Male-to-female	Male-to-male	Male-to-female	Male-to-male	Male-to-female	Male-to-male	Male-to-female	Male-to-male	Male-to-female
$t-1$	42.13%	41.58%	30.23%	26.81%	14.28%	15.22%	14.27%	15.91%	6,229	9,177
t	43.35%	42.87%	31.21%	27.70%	13.63%	14.77%	14.30%	15.83%	6,810	9,674
$t+1$	42.97%	43.56%	30.38%	26.58%	13.64%	14.24%	13.86%	14.89%	7,145	10,135
$t+2$	43.30%	42.29%	30.06%	25.35%	13.27%	14.23%	13.64%	15.24%	7,828	10,743

Panel B: Board and executive characteristics, and executive compensation.

Variables	Male executive		Female executive		Test of difference	
	Mean	Median	Mean	Median	t-statistic	z-statistic
Board Size	9.02	9.00	8.33	9.00	1.46	1.13
Board Independence (percent)	54.50	63.64	55.27	66.67	0.27	0.55
CEO/Chairman Duality (percent)	82.29	100.00	81.03	100.00	0.34	0.34
Female Representation on Board (percent)	10.22	10.00	14.08	12.50	3.30***	3.16***
MBA (percent)	48.57	0.00	56.00	100.00	1.01	1.01
Executive Age (years)	47.21	47.00	44.12	44.00	4.69***	3.83***
Hired from Inside (percent)	43.19	0.00	64.29	100.00	3.51***	7.37***
Total Compensation (thousands of dollars)	3,490.23	1,855.22	2,237.24	1,319.15	2.74***	3.56***
Cash Compensation (percent)	46.60	44.01	49.71	49.32	1.17	1.19
Stock and Option Compensation (percent)	36.46	34.52	36.60	35.05	0.05	0.05

years before and three years after an executive transition, excluding the year of the transition. We require that the firm

include firm fixed effects. Appendix B provides the definitions of the variables for our tests.

(b) Table 2: firm, board, executive, and compensation characteristics.

5.2 Tables 3–4: hiring female executives and decision outcomes

Read:

- Table 3 asks what predicts hiring a female executive. Market-to-book is positive and significant in several specifications; executive age is negative and significant; hired-from-inside is positive and significant.
- These patterns mean female executives are more likely to be promoted internally and tend to be younger.
- Table 4 is the main decision-outcome table. In Panel A, $Post \times Female$ is negative for asset growth, acquisitions, market leverage, and debt issuance.
- The acquisition coefficients in the main difference-in-differences design are about -0.050 and -0.045, indicating less acquisition activity after male-to-female transitions.

- The debt issuance coefficients are also negative and significant in the preferred specifications.
- Equity issuance does not show the same robust negative pattern.

Economic message: Female executives are associated with fewer large discretionary actions, especially acquisitions and debt issuance. This is the first major piece of evidence consistent with men being more overconfident in significant corporate decisions.

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Table 3

Probit regression of hiring a female executive. This table presents regression analysis of the likelihood of hiring a female executive. The dependent variable is a binary variable that equals one if a transition firm hires a female executive (Chief Executive Officer or Chief Financial Officer) and zero otherwise. See Appendix B for the definition of the independent variables. In regressions with industry dummies (Columns 2–4), we exclude industry code 12 (Others) in Fama and French 12 industry classifications. Numbers in parentheses are *t*-statistics based on Huber-White standard errors clustered by industry. Significance on a 10% (*), 5% (**), or 1% level (***) is indicated.

	(1)	(2)	(3)	(4)
Profitability	0.048 (1.00)	0.030 (0.62)	-0.021 (0.40)	0.026 (0.27)
Log(Size)	0.004 (0.82)	0.006 (1.30)	0.007 (1.20)	-0.026*** (3.08)
Market-to-Book	0.002 (1.45)	0.002** (2.08)	0.002** (2.20)	0.005*** (3.64)
PPE	-0.002 (0.28)	-0.003 (0.52)	-0.002 (0.34)	0.020 (1.61)
Firm Age	-0.000 (0.31)	-0.000 (0.14)	0.000 (0.27)	0.000 (0.35)
Board Independence			0.030 (0.77)	0.057 (1.05)
Board Size			-0.018 (1.56)	-0.017 (0.87)
CEO/Chairman Duality			-0.018 (1.37)	-0.047* (1.91)
MBA				0.017 (0.81)
Executive Age				-0.240*** (3.82)
Hired from Inside				0.047*** (3.73)
Consumer Non-durables	-0.008*** (3.67)	-0.012*** (4.14)		0.088*** (5.80)
Consumer Durables	-0.036*** (20.37)	-0.038*** (15.19)		0.070*** (6.40)
Manufacturing	-0.020*** (5.99)	-0.022*** (6.22)		0.010 (1.04)
Energy	-0.043*** (8.37)	-0.050*** (10.70)		
Chemicals	-0.031*** (11.74)	-0.030*** (11.54)		0.015** (2.34)
Business Equipment	-0.033*** (8.19)	-0.039*** (7.66)		0.024 (1.33)
Telecommunications	-0.015*** (4.41)	-0.034*** (6.97)		0.089*** (3.50)
Utilities	-0.018 (1.64)	-0.022** (2.21)		0.030 (1.14)
Shops	0.003** (2.30)	-0.003* (1.87)		0.051*** (6.44)
Health Care	-0.031*** (10.20)	-0.035*** (7.05)		0.043*** (2.82)

(a) Table 3: probit model for hiring female executives.

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Table 4

Asset growth and capital structure decisions. This table presents results on asset growth and capital structure decisions using three specifications. The dependent variables are the decision variables of interest, including asset growth, acquisitions, and capital structure decisions. Panel A reports difference-in-differences regression results from Eq. (1). Panel B reports the regression results using the full panel of firms. Panel C reports results from a propensity score matching approach. We first run a probit regression to pair each female-year observation with five male-year observations with statistically the same profitability, size, market-to-book ratio, plant, property, and equipment (PPE), year, industry membership, and an indicator for Chief Executive Officers. We then run a univariate regression of the decision variables of interest on the female dummy. See Appendix B for the definition of the independent variables. We winsorize all financial ratios at the 2.5% level to reduce the effect of outliers. Numbers in parentheses are *t*-statistics based on Huber-White standard errors. Significance on a 10% (*), 5% (**), or 1% level (***) is indicated.

Panel A: Difference-in-difference regressions.										
	Asset Growth		Acquisitions		Leverage		Debt Issuance		Equity Issuance	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Post × Female	-0.025** (2.29)	-0.016* (1.69)	-0.050** (2.28)	-0.045* (1.91)	-0.018** (2.03)	-0.010 (1.55)	-0.335** (2.10)	-0.280* (1.88)	0.169* (1.84)	0.074 (0.74)
Post	-0.029*** (4.45)	-0.009 (1.42)	-0.031*** (2.81)	-0.005 (0.51)	-0.020*** (4.41)	-0.006** (2.25)	-0.124** (2.36)	0.000 (0.00)	0.037 (0.81)	0.007 (0.16)
Profitability	-0.126** (2.47)	0.619*** (7.49)	0.193** (2.24)	0.671*** (5.50)	-0.647*** (22.36)	-0.379*** (11.62)	-1.442*** (3.66)	0.961* (1.70)	-2.663*** (8.98)	-0.912** (2.02)
Log(Size)	-0.003 (0.69)	-0.030* (1.76)	0.161*** (17.48)	0.127*** (7.31)	-0.072*** (16.74)	0.000 (0.06)	0.792*** (21.37)	0.537*** (5.99)	0.995*** (26.76)	0.627*** (8.35)
Market-to-Book	0.058*** (14.59)	0.078*** (8.60)	-0.086*** (11.11)	-0.054*** (4.73)	-0.030*** (10.38)	-0.021*** (8.64)	-0.395*** (10.88)	-0.115*** (2.04)	0.108*** (4.29)	0.116*** (3.16)
PPE	-0.021*** (5.59)	-0.137*** (6.95)	-0.198*** (22.21)	-0.243*** (11.34)	0.082*** (22.94)	0.037*** (7.09)	-0.050 (1.58)	0.072 (0.81)	-0.114*** (3.77)	-0.198*** (2.89)
CapEx	0.633*** (9.77)	0.725*** (5.06)	-0.347*** (3.83)	0.328** (2.43)	-0.577*** (11.22)	0.080* (1.66)	2.555*** (5.56)	2.692*** (3.77)	2.304*** (4.30)	-0.027 (0.04)
Firm fixed effects	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	10,183	10,183	9,357	9,357	10,180	10,180	7,206	7,206	8,261	8,261
Adj. R-squared	0.15	0.33	0.17	0.33	0.41	0.80	0.25	0.48	0.39	0.57
Panel B: Panel regressions.										
	Asset Growth		Acquisitions		Leverage		Debt Issuance		Equity Issuance	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Female	-0.032*** (3.09)	-0.014 (1.16)	-0.050** (2.24)	-0.043* (1.68)	-0.013 (1.53)	-0.013** (2.31)	-0.282** (2.00)	-0.282** (2.05)	0.097 (1.08)	0.018 (0.19)
Profitability	0.004 (0.08)	0.615*** (8.12)	0.293*** (3.07)	0.806*** (5.44)	-0.677*** (20.96)	-0.343*** (9.52)	-1.989*** (4.58)	2.014*** (2.95)	-1.548*** (4.77)	-0.154 (0.33)
Log(Size)	0.013*** (3.12)	-0.092*** (7.15)	0.187*** (17.84)	0.089*** (2.86)	-0.072*** (19.59)	-0.005 (0.71)	0.818*** (20.47)	0.392*** (3.44)	1.077*** (32.25)	0.620*** (7.84)
Market-to-Book	0.044*** (9.80)	0.110*** (13.44)	-0.100*** (9.95)	-0.028 (1.51)	-0.042*** (13.69)	-0.024*** (6.94)	0.412*** (9.42)	-0.054 (0.44)	0.031 (1.13)	0.086* (1.95)
PPE	-0.023*** (3.09)	-0.122*** (11.6)	-0.211*** (17.8)	-0.303*** (13.6)	0.078*** (13.6)	0.036** (2.0)	-0.076* (1.8)	-0.105*** (2.6)	-0.269*** (3.1)	

(b) Table 4: gender and corporate decision outcomes.

5.3 Table 5: announcement returns for acquisitions and security issuance

Read:

- The dependent variable is the three-day announcement CAR.
- For acquisitions, $Post \times Female$ is 2.114% using raw CARs and 1.858% using market-adjusted CARs; both are statistically significant.
- For debt issuance, $Post \times Female$ is 5.447% raw and 3.693% market-adjusted; both are positive and statistically significant or close to significant.
- For equity issuance, $Post \times Female$ is not statistically significant.
- The stronger market reaction for acquisitions and debt issuance is consistent with the idea that female executives impose a stricter threshold for major corporate actions.

Economic message: The table links decision frequency to shareholder value. It is not only that female executives do fewer acquisitions and debt issues; the transactions they do undertake are received more favorably by the market.

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Table 5

Announcement returns for acquisitions, debt issuance, and equity issuance.
This table presents regression results from Eq. (2). The dependent variable is the three-day cumulative abnormal announcement return (CAR) around announcements of acquisitions, equity issuance, and debt issuance. We consider four years surrounding the year of the executive transition, i.e., two years before the transition and two years after. *Toehold* is a dummy that equals one if the acquirer holds a minority interest position (less than 50%) in the target's stock before the announcement. *Stock Deal* is a dummy variable that equals one if the fraction of equity value as a payment method in the deal exceeds 50%. *Public Target* is a dummy variable that equals one if the target is a publicly listed firm. *Diversifying* is a dummy variable that equals one if the acquirer and the target are not from the same Standard Industrial Classification industry. *Hostile*, *Contested*, *Tender*, and *Merger of Equals* are dummy variables that equal one if the takeover is unsolicited or unfriendly, has multiple bidders, is a tender offer, and is classified as merger of equals by Securities Data Company, respectively. *Return Volatility* is the standard deviation of daily stock return during the trading period (−90, −11) prior to the announcement date (trading day 0). *Share Turnover* is the ratio of average daily share trading volume during the trading period (−90, −11) prior to the announcement date (trading day 0) divided by pre-issuance total shares outstanding. See Appendix B for the definition of other variables. Numbers in parentheses are *t*-statistics based on Huber-White standard errors clustered by industry. Significance on a 10% (*), 5% (**), or 1% level (***) is indicated.

	Acquisition CARs		Debt Issuance CARs		Equity Issuance CARs	
	Raw	Market-adjusted	Raw	Market-adjusted	Raw	Market-adjusted
	(1)	(2)	(3)	(4)	(5)	(6)
<i>PostFemale</i>	2.114** (2.61)	1.858** (2.22)	5.447** (2.19)	3.693* (1.95)	−0.377 (0.27)	0.415 (0.32)
<i>Post</i>	−0.120 (0.27)	−0.140 (0.34)	−0.596 (0.41)	−0.323 (0.30)	0.243 (0.50)	0.054 (0.11)
<i>Female</i>	0.572 (1.17)	0.426 (0.85)	−0.361 (0.26)	−1.232 (1.12)	0.590 (0.45)	0.754 (0.70)
<i>LogSize</i>	−0.324** (2.24)	−0.340*** (2.68)	−1.710 (1.03)	−0.531 (0.45)	0.073 (0.35)	0.579 (0.22)
<i>Market-to-book</i>	0.078** (2.18)	0.058* (1.73)	0.053 (0.05)	−0.362 (0.46)	0.201** (2.25)	0.196** (2.28)
<i>Cash Flow</i>	0.134 (0.24)	0.055 (0.08)	0.631 (0.03)	13.082 (0.93)	−25.851** (2.47)	−21.371** (2.33)
<i>Sales Growth</i>	−0.145 (0.78)	−0.116 (0.60)	0.002 (0.00)	−0.274 (0.16)	−0.134 (0.67)	−0.152 (0.93)
<i>Hostile</i>	−1.034 (0.69)	−0.826 (0.54)				
<i>Toehold</i>	−1.856 (0.76)	−2.335 (1.03)				
<i>Stock Deal</i>	0.029 (0.04)	0.050 (0.07)				
<i>Public Target</i>	−1.981*** (4.51)	−2.467*** (5.79)				
<i>Diversifying</i>	−0.107 (0.25)	−0.148 (0.34)				
<i>Contested</i>	0.053 (0.63)	0.370 (0.31)				
<i>Tender</i>	1.986** (2.46)	2.279** (3.74)				
<i>Merger of Equals</i>	1.842 (1.44)	1.517 (1.22)				
<i>Profitability</i>			−0.207 (0.64)	−0.298 (1.33)	9.061 (1.11)	9.479 (1.31)
<i>PPE</i>			1.334 (1.00)	0.063 (0.07)	0.252 (0.94)	0.320 (1.23)
<i>Return Volatility</i>			19.369 (0.10)	82.802 (0.06)	−16.969 (0.51)	−24.578 (0.79)
<i>Share Turnover</i>			2.360 (1.22)	1.097 (0.66)	−1.504** (2.14)	−1.473** (2.20)
<i>LogNet Proceeds</i>			0.361 (0.77)	−0.024 (0.07)	−0.138 (0.40)	−0.046 (0.16)
Year-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	2,783	2,783	111	111	592	592
Adj. R-squared	0.02	0.03	0.23	0.34	0.13	0.10

(footnote continued)
specification for other tests of the paper, the results of which are not reported due to space concerns. Most of the other results of the paper are robust to using this alternate empirical design, with the

(footnote continued)
not robust for debt issuance CARs, this is due to a lack of significance of the instrument in the first stage on what is a significantly reduced sample in this test (only 111 public debt issuance observations during

5.4 Tables 6–7: instrumental variables and forecast precision

Read:

- Table 6 uses state-level gender equality as an instrument for having a female executive.
- The first-stage relation is positive and significant: higher gender equality is associated with a higher probability of a female executive.
- The second-stage estimates support the main results: instrumented female executive presence is associated with fewer acquisitions and higher acquisition announcement returns.
- The first-stage F-statistics are around 8.2–8.4, so the authors caution that weak-instrument concerns cannot be ruled out entirely.
- Table 7 studies earnings forecast spreads. For male-to-female transition firms, forecast spreads rise from 0.035 before transition to 0.113 after transition.
- For male-to-male transition firms, forecast spreads rise only from 0.034 to 0.049.
- The difference-in-differences estimate is 0.063 and significant at the 1% level.

Economic message: The IV table is a robustness check against selection. The forecast-spread table is a direct test of overprecision: male executives issue narrower forecasts, while female executives place wider bounds around earnings estimates.

Table 6

Acquisitions and acquisition announcement returns: instrumental variable approach.

This table presents the two-stage least squares regression results from Eq. (3). Columns 1 and 3 report the results from the first-stage ordinary least squares regressions with the female dummy as the dependent variable. *F*-statistics and Wald test statistics from the first-stage regressions are reported at the bottom of the table. Columns 2 and 4 report the results for the second-stage regressions with acquisition and acquisition announcement returns as the dependent variables. *Instrumented Female* is the fitted value of the female indicator from the first-stage regression. *Gender Equality* is the state-level gender equality index proposed by Sugarman and Straus (1988). See Appendix B for the definition of other variables. Numbers in parentheses are *t*-statistics based on Huber-White standard errors. Significance on a 10% (*), 5% (**), or 1% level (***) is indicated.

	Acquisitions		Acquisition CARs (percent)	
	First stage	Second stage	First stage	Second stage
	(1)	(2)	(3)	(4)
<i>Instrumented Female</i>		−3.178** (2.35)		24.541** (2.39)
<i>Profitability</i>	0.041 (1.46)	0.431*** (3.33)	0.017 (0.26)	3.432* (1.76)
<i>Log(Size)</i>	0.006** (2.47)	0.206*** (13.83)	0.006 (1.02)	−0.166 (0.85)
<i>Market-to-Book</i>	−0.001 (0.40)	−0.101*** (8.79)	−0.007 (1.31)	0.053 (0.30)
<i>PPE</i>	−0.002 (1.01)	−0.220*** (19.62)	0.000 (0.07)	0.033 (0.22)
<i>Hostile</i>			−0.016 (0.33)	−0.083 (0.06)
<i>Threshold</i>			−0.003	−1.979***

(a) Table 6: IV approach using state-level gender equality.

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cluding point forecasts from the sample does not change results). Table 7 presents difference-in-differences tests management forecasts of EPS. This test indicates that ns with female executives are more likely to give a wider ecast range than those with male executives. The first row ws that the forecast spread is 11.3% for the two years er a female executive takes over, compared with 3.5% for two years prior to the female executive taking over. en the difference-in-differences is taken, we find that ale executives make EPS forecasts that are 6.3% wider n male executives, and this result is significant at 1%. ese results provide direct support that male executives are rconfident relative to women. In unreported tests, we do

Table 7

nings forecasts.

his table presents univariate difference-in-differences results on the ad of management earnings per share forecasts. We consider four rs surrounding the year of the executive transition, i.e., two years re the transition and two years after. The spread of management EPS casts is defined as the difference between the lower bound and the er bound of the forecast range divided by the midpoint of the cast range. In the case of point forecasts, the forecast spread is zero. here in parentheses are *t*-statistics and numbers in square brackets

(b) Table 7: management earnings forecast spreads.

5.5 Tables 8–9: career outcomes, option behavior, and value-destroying acquisitions

Read:

- Table 8 Panel A shows female CFOs are more likely to stay at least four years: 51.12% versus 44.16% for male CFOs. The difference is statistically significant.
- Female CEOs also have a higher four-year retention rate, 63.89% versus 54.83%, though the difference is not statistically significant.
- Table 8 Panel B replicates Malmendier and Tate-style overconfidence measures. Male executives have higher Holder 67, Long Holder, and Net Buyer values.
- Holder 67 is 31.29% for males and 18.18% for females; Long Holder is 33.21% for males and 19.61% for females. Both differences are significant.
- Table 9 shows male-led acquisitions are more likely to be value destroying. In male-to-female transition firms, the share of negative-CAR acquisitions falls from 50.00% before the transition to 29.31% after the female executive takes over.
- In male-to-male transition firms, the value-destroying share remains essentially unchanged: 50.51% before versus 49.59% after.
- The same pattern holds for cash-only acquisitions, which addresses the concern that stock-financed acquisitions mechanically generate worse announcement returns.

Economic message: These tests are central to the paper's interpretation. Risk aversion could explain fewer actions, but it does not predict narrower male forecasts, delayed option exercise, worse male career outcomes, and a higher male probability of value-destroying acquisitions. The combined evidence points to

relative male overconfidence.

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Table 8

Career outcomes and stock option exercises.

This table reports results on career outcomes of executives behaviors. Panel A reports the probability of an executive to rer reports the proportion of executives that exhibit certain option the measures proposed by Malmendier and Tate (2005). In p exercise his options in the first five years after vesting althoug during the five years, a Long Holder holds an option until the buyer of his company's stock during the first three years of his t (***) is indicated.

Panel A: Employment outcomes.

Prob(Staying for at least four years as a Chief Executive Offi
Prob(Staying for at least four years as a Chief Financial Offi

Panel B: Option exercising and stock trading behavior.

	Male	Female
Holder 67	31.29%	18.18%
Long Holder	33.21%	19.61%
Net Buyer	38.90%	31.37%

(a) Table 8: career outcomes and option-exercise measures.

Panel A of Table 9 reports a difference-in-differences test for the full sample of acquisitions. Among male-to-female transition firms, over half of acquisitions were value destroying (with negative announcement returns) when the firm had a male executive. After a female executive takes over, only about one-third of acquisitions are value destroying. Among male-to-male transition firms, the likelihood for value destroying acquisitions is similar before and after a transition. Because acquisitions financed with equity generally have worse announcement returns than those financed with cash, we conduct the same test but with acquisitions funded by cash only. For this sample, the results are similar, indicating that the

Table 9

Value-destroying acquisitions.

This table presents results on the probability of value-destroying acquisitions (i.e., acquisitions with negative abnormal returns around announcement). We consider four years surrounding the year of the executive transition, i.e., two years before the transition and two years after. Panels A and B report the univariate difference-in-differences result for all acquisitions and the subsample of cash offers, respectively. Numbers in parentheses are *t*-statistics, and numbers in square brackets are the numbers of observations for each subsample. Significance on a 10% (*), 5% (**), or 1% level (***) is indicated.

Panel A: All acquisitions.

	Before	After	Difference (after-before)
Male-to-female	50.00%*** (9.22) [86]	29.31% (4.86)*** [58]	-20.69%** (2.51)
Male-to-male	50.51%*** (37.59) [1,386]	49.59%*** (36.53) [1,357]	-0.91% (0.48)
Difference (female-male)	-0.51% (0.09)	-20.28%*** (3.03)	-19.78%** (2.27)

(b) Table 9: probability of value-destroying acquisitions.

5.6 Appendix A: male-to-female executive transitions

Read:

- Appendix Table A1 lists the male-to-female executive transitions used in the sample.
- For each transition, it reports company name, succeeding executive name, position, start year, tenure, predecessor name, and industry classification.
- The list includes both CEOs and CFOs, with CFOs being more numerous.
- The appendix confirms the hand-collected nature of the sample and helps make the transition design transparent.

Economic message: The appendix matters because the paper's credibility depends heavily on careful identification of executive gender and transition timing. The table documents the treated transitions rather than leaving the treatment group opaque.

Table A1

Company name	Executive name	Position	Start	Tenure	Predecessor name	Industry classification
AGL Resources Inc.	Paula G. Kropp	CEO	2000	6	Walter M. Higgins	Utilities
Alaska Communications Systems	Liane Pelletier	CEO	2003	3	Charles E. Robinson	Communication
Albertson's Inc.	Felicia D. Thornton	CFO	2001	5	A. Craig Olson	Retail
Alpharma Inc.	Ingrid Wolk	CEO	1999	7	Elmar W. Kissner	Pharmaceutical products
American Electric Power	Susan Tomasky	CEO	2001	5	Henry W. Fayne	Utilities
American Water Works Inc.	Ellen C. Wolf	CFO	1999	4	J. James Barr	Utilities
Amgen Inc.	Kathryn E. Falberg	CEO	1998	3	Robert S. Artyeh	Pharmaceutical products
Applied Materials Inc.	Nancy H. Handel	CEO	2002	4	Joseph R. Bronson	Machinery
Atlantic Richfield Co.	Marie L. Goodles	CFO	1996	5	Ronald J. Anhalt	Petroleum and natural gas
Automatic Data Processing	Karen E. Dykstra	CFO	2001	5	Richard J. Haviland	Business services
Avon Products	Andrea Jung	CEO	1999	7	Charles R. Perrin	Consumer goods
Avicel Technologies Inc.	Mary G. Puma	CEO	2001	5	Brian R. Bachman	Machinery
Banta Corp.	Stephanie A. Streeter	CEO	2002	4	Donald D. Belcher	Business services
Barnes & Noble Inc.	Irene R. Miller	CEO	1994	4	Michelle Clipper	Retail
Belo Corp.	Dunia A. Shive	CFO	1998	4	Michael D. Perry	Printing and publishing
Benchmark Electronics Inc.	Gayla J. Delly	CFO	2001	5	Cary T. Fu	Electronic equipment
Bio-Rad Laboratories Inc.	Christine A. Tsingos	CFO	2002	4	Norman Schwartz	Measuring and control equipment
Brown-Forman Corp.	Phoebe A. Wood	CFO	2000	6	Steven B. Ratoff	Beer and liquor
Brunswick Corp.	Victoria J. Birch	CFO	1999	4	Peter R. Hamilton	Shipbuilding, railroad equipment
Cardinal Companies Inc.	Carol P. Lowe	CFO	2002	4	Kirk F. Vincent	Utilities
CDW Corp.	Barbara A. Klein	CFO	2001	5	Harry J. Harczak Jr.	Retail
Central Vermont Public Service	Joan H. Gibson	CFO	2002	4	Francis J. Boyle	Utilities
Champion Enterprises Inc.	Phyllis A. Knight	CFO	2002	4	Anthony S. Cleberg	Construction materials
Charming Shoppes Inc.	Dorris J. Bern	CEO	1995	11	David V. Wachs	Retail
Claire's Stores Inc.	Maria L. Schaefer	CEO	2002	4	Rowland Schaefer	Retail
Cleco Corp.	Dilek Samil	CFO	2001	4	Thomas J. Howlin	Utilities
Cleveland-Cliffs Inc.	Cynthia B. Bezik	CFO	1998	7	John S. Brizzo	Mining
Cleco Co.	Karen M. Rose	CFO	1998	6	William F. Ansfahl	Consumer goods
Coherent Inc.	Helene Simonet	CFO	2002	4	Robert J. Quillinan	Measuring and control equipment
Compuserve Corp.	Laura L. Fournier	CFO	1997	9	Ralph A. Caponigro	Business services
Consolidated Edison Inc.	Joan S. Freilich	CEO	1997	8	Raymond J. McCann	Utilities
Constellation Energy Group Inc.	E. Follis Smith	CEO	2001	4	David A. Hadden	Utilities
Corn Products International Inc.	Cheryl K. Beebe	CFO	2003	3	James W. Ripley	Food products
Cree Inc.	Cynthia B. Merrill	CEO	2001	8	Alan J. Robertson	Electronic equipment
Cummins Inc.	Joan S. Blackwell	CEO	2002	4	Tom Liebharger	Machinery
Cymer Inc.	Nancy J. Baker	CEO	2002	4	William A. Angus III	Machinery
Darden Restaurants Inc.	Linda J. Dimondos	CEO	2002	4	Clarence Olin	Restaurants, hotels, motels
Delta Air Lines Inc.	M. Michele Burns	CEO	2000	4	Edward H. West	Transportation
DirectTV Group Inc.	Roxanne S. Austin	CEO	1997	4	Charles Nooki	Communication
Dunnellay (R & R) & Sons Co.	Cheryl C. Francis	CEO	1995	5	Frank R. Jarc	Business services
DPL Inc.	Elizabeth M. McCarthy	CEO	2000	4	James P. Torgerson	Utilities
Dun & Bradstreet Corp.	Sara Mathew	CFO	2001	5	Chester J. Gjevda Jr.	Business services
Federal Signal Corp.	Stephanie K. Kushner	CEO	2002	4	Henry L. Dykema	Automobiles and trucks
First Data Corp.	Kimberly S. Palmore	CEO	1999	7	Lee Adrean	Business services
Fluor Corp.	Renee J. Horthaker	CEO	1997	8	Bruce E. Hines	Machinery
Freeport-McMoran Copper and Gold	Kathleen L. Quirk	CFO	2003	3	Stephen M. Jones	Mining
Gannett Co.	Linda J. Dimondos	CEO	2002	4	Harry F. McArthur	Printing and publishing
Gap Inc.	Heidi Kunz	CEO	1999	3	Warren R. Hathagen	Retail
Gartner Inc.	Regina M. Paulillo	CEO	1999	4	John F. Halligan	Business services
GenCorp Inc.	Yasmin K. Seyal	CEO	2001	5	Terry L. Hall	Business services
General Electric Co.	Keith S. Sherin	CEO	1998	8	Dennis D.	Defense
Getty Images Inc.	Sara Mathew	CEO	2000	6	Dannerman	Business services
Graybar Electric Co. Inc.	Janita H. Hineshaw	CEO	2000	5	Christopher J. Roling	Wholesale
Great Plains Energy Inc.	Andrea F. Biebler	CEO	1996	10	John DeDefraco	Utilities
Hewlett-Packard Co.	Carlton S. Fiorina	CEO	1999	6	Lewis E. Platt	Computers
Home Depot Inc.	Carol B. Tome	CEO	2001	5	Dennis J. Carey	Retail
Houghton Mifflin Co.	Gail Began	CEO	1995	6	Stephen O. Jaeger	Printing and publishing
HMS Health Inc.	Nancy E. Cooper	CEO	2001	5	James C. Malone	Business services
Inglis Markets Inc.	Brenda S. Tudor	CEO	1998	7	Jack R. Ferguson	Retail
International Game Technology	Maureen T. Mullarky	CEO	1998	8	David P. Hanlon	Business services
International Paper Co.	Marianne M. Parrs	CEO	1995	6	Robert C. Butler	Business supplies
International Speedway Corp.	Susan C. Schandel	CEO	1999	7	H. Lee Condit	Entertainment
ITT Corp.	Heidi Kunz	CEO	1995	4	Robert A. Bowman	Electronic equipment
Lee Enterprises Inc.	Mary E. Junck	CEO	2001	5	Richard D. Gottlieb	Printing and publishing
Lennox International Inc.	Michael A. Smith	CEO	2003	3	Richard A. Smith	Machinery
Limited Brands Inc.	V. Ann Hailey	CEO	1997	9	Kenneth R. Gilman	Retail

(a) Appendix Table A1, excerpt 1.

Table A1 (continued)

Company name	Executive name	Position	Start	Tenure	Predecessor name	Industry classification
Lucent Technologies Inc.	Patricia F. Russo	CEO	2002	4	Henry R. Schacht	Computers
May's Inc.	Karen M. Hargett	CFO	1995	11	Ronald W. Tyoe	Retail
Master Inc.	Carmen M. Sabater	CFO	1998	4	Edwin D. Johnson	Construction
Mattel Inc.	Francesca Lazuriga	CFO	1994	3	Michael C. McCafferty	Toys recreation
Medimmune Inc.	Lois S. Zoff	CFO	2003	3	Gregory S. Patrick	Pharmaceutical products
Miller (Herman) Inc.	Elizabeth A. Nickels	CFO	1999	6	Brian C. Walker	Business supplies
Millipore Corp.	Kathleen B. Allen	CFO	1999	7	Francis J. Langer	Measuring and control equipment
New York Times Co.	Diane P. Baker	CFO	1995	3	David L. Gorham	Printing and publishing
Nicor Inc.	Kathleen L. Halloran	CFO	1999	4	David L. Gyzanski	Utilities
NL Industries	Susan E. Alderton	CFO	1997	4	Joseph S. Compocicelli	Machinery
Oakwood Homes Corp.	Suzanne H. Wood	CFO	2000	3	Robert A. Smith	Construction materials
OMI Corp.	Kathleen C. Haines	CFO	2000	6	Vincent J. de Souza	Transportation
Owens & Minor Inc.	Ann Greer Rector	CFO	1996	3	Glenn J. Dozier	Wholesale
Pathmark Stores Inc.	Ellen R. Scott	CEO	2002	3	James Donald	Retail
Pepsico Inc.	Indra K. Nooyi	CFO	2002	4	Matthew M. McKenna	Beer and liquor
Pharmaceutical Product Development Inc.	Linda Baddour	CFO	2002	4	Philippe M. Maitre	Business services
Plum Creek Timber Co. Inc.	Diane M. Irvine	CFO	1995	4	Rick R. Holley	Construction materials
Portland General Electric Co.	Peggy V. Fowler	CEO	2000	6	Ken L. Harrison	Utilities
Range Resources Corp.	Edwin M. Lefflane III	CFO	1999	4	Thomas W. Stoelk	Petroleum and natural gas
Reliance Steel & Aluminum Co.	Karla R. McDowell	CFO	1998	8	Steven S. Weiss	Wholesale
Reynolds American Inc.	Dianne M. Neal	CFO	2001	5	Kenneth J. Lapijko	Tobacco products
Rite Aid Corp.	Mary F. Sammons	CEO	2003	3	Robert C. Miller	Retail
Ruby Tuesday Inc.	Marguerite N. Duffy	CFO	2001	4	J. Russell Mothershead	Restaurants, hotels, motels
Ryder System Inc.	Taryn A. Leimbach	CFO	2003	3	Corlis J. Nelson	Personal services
Sara Lee Corp.	Judith A. Sprieser	CEO	1995	5	Michael E. Murphy	Food products
Smith International Inc.	Margaret K. Dorman	CFO	1999	7	John J. Kennedy	Chemicals
Spherulite	Stephen O. Jaeger	CEO	1994	5	Michael L. Ainslie	Business services
Sphertion Corp.	Cinda A. Hallman	CEO	2001	3	Raymond Marcy	Business services
Steris Corp.	Laurie Blas	CFO	1999	7	Mark L. Fagerholm	Medical equipment
Sunoco Logistics Partners L.P.	Deborah M. Fretz	CEO	2001	5	John G. Drouck	Transportation
Supervalu Inc.	Pamela K. Knous	CFO	1997	9	Jeffrey C. Girard	Retail
Telephone & Data Systems Inc.	Sandra L. Helton	CFO	1998	7	Murray L. Swanson	Communication
TeleTech Holdings Inc.	Margot O'Brien	CFO	2000	3	Michael E. Foss	Business services
Tollaba Inc.	Joan E. Ryan	CFO	2000	3	Peter A. Guglielmi	Electronic equipment
Unisys Corp.	Janet R. Haugen	CFO	1999	7	Robert H. Brust	Computers
United States Steel Corp.	Gretchen R. Haggerty	CFO	2002	4	John P. Surma	Steel works, etc.
USEC Inc.	Ellen C. Wolf	CFO	2003	3	Henry Z. Shelton Jr.	Chemicals
Vianor Medical Systems Inc.	Elisha W. Finney	CFO	1999	7	Robert S. Lemus	Medical equipment

(b) Appendix Table A1, excerpt 2.

6 Section-by-section walkthrough

6.1 Section 1: Introduction

The introduction frames the paper around the idea that traditional corporate finance often abstracts from managerial identity. The authors argue that specific managers matter, and gender is one potentially important characteristic. They emphasize that female representation among top executives increased from the mid-1990s to the mid-2000s but remained low, creating a natural setting for studying whether female executives make different choices.

The introduction also previews the central findings: female executives make fewer acquisitions, issue debt less often, and receive more favorable market reactions when they do make acquisitions or debt offerings. The interpretation is not simply that women are more conservative. The authors argue that the pattern fits male overconfidence because male executives appear to undertake more value-destroying actions.

6.2 Section 2.1: Gender behavioral differences

This subsection translates the psychology literature on overconfidence into corporate finance predictions. An overconfident manager overestimates payoff quality or underestimates risk. That executive will accept projects that a less overconfident manager rejects. This generates two testable predictions: male executives should act more often, and the marginal male-led actions should perform worse.

The subsection also distinguishes overconfidence from risk aversion. Female risk aversion and male overconfidence both predict fewer actions by women, but they differ in their implications for value and careers. If female risk aversion leads women to avoid profitable projects, female executives should eventually have worse career outcomes. If male overconfidence leads men to accept bad projects, male executives should have worse career outcomes and male-led acquisitions should be more often value destroying.

6.3 Section 2.2: Significance of CEOs and CFOs

The paper justifies including both CEOs and CFOs. Prior work shows that CEOs and CFOs have manager-specific effects on firm policies. CFOs, in particular, are closely tied to financing decisions and acquisition execution. Since the sample contains many more female CFOs than female CEOs, excluding CFOs would reduce statistical power and omit an executive role that is directly relevant to the outcomes.

6.4 Section 3.1: Empirical methodology

This subsection discusses why the main empirical problem is selection. Female executives may work at different firms, boards may hire female executives for particular reasons, and female executives may be more common in some industries. The authors respond with a transition design, multiple controls, fixed effects, and robustness checks.

The main identifying idea is to compare a firm's behavior before and after an executive transition. Male-to-female transitions are compared with male-to-male transitions, so the control group captures the normal effect of executive turnover. The authors also exclude the transition year to avoid ambiguity about which executive made decisions during that year.

6.5 Section 3.2: Summary statistics

The summary statistics show that female executives are not randomly distributed. Female executives are more common in later years and in consumer industries, are younger, are more likely to be hired internally, and work at larger firms. These differences make the empirical design essential. They also provide hints about the labor market for female executives: internal promotion may be more common because internal female candidates can better demonstrate ability in the presence of external discrimination or information frictions.

6.6 Section 3.3: Managerial decision making

The main decision results show that female executives make fewer major corporate decisions. Asset growth is lower after male-to-female transitions, and acquisition activity falls. Debt issuance also falls. Leverage falls in some specifications, but the evidence is less uniform than for acquisitions and debt issuance. Equity issuance is not robustly different.

The interpretation is that overconfident male executives are more willing to undertake discretionary actions. Acquisitions are especially important because they require managerial judgment and often reflect executive confidence in evaluating and integrating another firm. Debt issuance matters because it can finance expansion and exposes the firm to financial risk.

6.7 Section 3.4: Announcement returns

The announcement-return tests ask whether fewer actions by female executives are better or worse for shareholders. The evidence is favorable to female executives: acquisitions and debt issues receive higher market reactions when undertaken by female-led firms after male-to-female transitions. This is crucial because it separates cautiousness from quality.

A pure risk-aversion story could imply that women avoid some positive-NPV deals, which would hurt shareholders. But the announcement-return evidence suggests that female-led transactions are not worse; they are better received by investors. This supports the view that female executives reject marginal deals that investors dislike.

6.8 Section 3.5: Instrumental variable approach

The IV analysis uses state-level gender equality as an instrument. The idea is that firms headquartered in states more favorable to women's equality are more likely to have female executives. The first stage is significant, and the second stage confirms fewer acquisitions and higher acquisition CARs for firms with female executives.

The authors are cautious because the first-stage F-statistics are below 10, meaning the instrument is not extremely strong. Still, the IV evidence is directionally consistent with the main DiD analysis and helps address lingering concerns about firm-level selection.

6.9 Section 4.1: Earnings forecasts

The earnings forecast test provides a direct measure of overprecision. Overconfident executives should provide tighter ranges around earnings forecasts. The paper finds that after female executives take over, forecast spreads become much wider relative to male-to-male transitions. This is hard to explain by risk aversion alone and directly supports the overconfidence mechanism.

6.10 Section 4.2: Career outcomes

Career outcomes test whether male or female executives are penalized more often. If women were simply too risk averse and therefore harmed shareholders by avoiding profitable actions, female executives should have worse career outcomes. Instead, female CFOs are more likely than male CFOs to remain in office at least four years, and the CEO difference is also directionally favorable to women. This supports the view that male overconfidence imposes costs.

6.11 Section 4.3: Stock option exercise and stock trading

The option-exercise tests use Malmendier and Tate-style measures. Overconfident executives believe their firm's stock is undervalued or expect future performance to be better than it is, so they delay exercising deep-in-the-money options and sometimes buy additional company stock despite already being highly exposed. Male executives score higher on these measures, again supporting the overconfidence interpretation.

6.12 Section 4.4: Value-destroying acquisitions

This subsection is the cleanest shareholder-value test. Male executives do not just undertake more acquisitions; they undertake a higher fraction of acquisitions with negative announcement returns. In the treated group, the negative-CAR acquisition share falls sharply after a female executive takes over. The same pattern appears for cash-only deals, reducing concerns that the result is driven by stock-financing effects.

6.13 Section 5: Conclusion

The conclusion emphasizes that executive gender is associated with differences in corporate financial and investment decisions. Female executives are less likely to make acquisitions and issue debt, and their major decisions receive more favorable market reactions. The paper interprets the breadth of evidence as most consistent with male executives being overconfident relative to female executives.

7 Short summary

- The paper studies whether executive gender affects corporate finance decisions.
- The key hypothesis is that male executives are more overconfident than female executives.
- The sample contains 1,750 male and 116 female executives, with transitions from 1994–2003.
- The main design compares male-to-female executive transitions with male-to-male transitions.
- Female executives are more likely to be hired later in the sample, in consumer industries, from inside the firm, and at younger ages.
- After male-to-female transitions, firms grow more slowly, make fewer acquisitions, and issue debt less often.
- Acquisition announcement returns are roughly 2 percentage points higher under female executives.

- Debt issuance announcement returns are also higher under female executives.
- Earnings forecasts issued under female executives have wider ranges, consistent with lower overprecision.
- Female CFOs have better career retention outcomes than male CFOs.
- Male executives are more likely to hold deep-in-the-money options and hold options until expiration.
- Male executives are more likely to undertake acquisitions with negative announcement returns.
- The IV evidence using state-level gender equality supports the acquisition and announcement-return results, though the authors caution about instrument strength.
- The combined evidence is most consistent with relative male overconfidence, rather than only female risk aversion.

8 Conclusion

8.1 Core conclusion

Huang and Kisgen show that firms behave differently after hiring female executives. They make fewer acquisitions, issue debt less often, and receive more favorable investor reactions to major transactions. The pattern is economically meaningful and robust across several empirical designs.

8.2 Economic intuition

Overconfident executives think more projects are worth doing than actually are. If men are relatively more overconfident, male executives will undertake more actions, but some of those actions will be value destroying. Female executives appear to impose a higher threshold for action. That leads to fewer acquisitions and debt issues, but better average market reactions when the firm does act.

8.3 Takeaway

The paper's larger message is that corporate finance decisions are shaped by who runs the firm, not only by firm-level characteristics. Executive gender is one observable dimension of managerial heterogeneity that helps explain acquisitions, debt issuance, announcement returns, and managerial overconfidence.